MAHATMA GANDHI UNIVERSITY , KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER PRACTICAL EXAMINATION

Minor- DSC B: MG3DSCPHY202: Atomic and Molecular Spectroscopy

Duration: 2 hours Maximum Marks: 35

Record: 5 Marks

Attempt the question marked 'X' . Change of question is not allowed. (30 Marks).

(Do error calculations wherever necessary)

1. Verify Beer Lambert's law by varying the concentration of the given liquid. [U, A, An] [6]
2. By changing path length, prove Beer Lambert's law. [U, A, An] [6]
3. Determine the absorption coefficient of the given solution at a particular wavelength by using a laser and a detector. [U, A, An] [6]
4. An equilateral prism is given. Determine the refractive index of the material of the prism using a light source and spectrometer.
   [U, A, An] [6]
5. Determine the dispersive power of the given prism using two prominent wavelengths. [U, A, An] [6]
6. Determine the dispersive power of the given grating using two prominent wavelengths.
   [U, A, An] [6]
7. Determine the wavelengths of prominent lines in the mercury spectrum using grating and spectrometer. [U, A, An] [6]
8. Using a suitable computational software obtain the vibrational frequencies, bond length, bond angle, dipole moment & Total energy of H_2O molecule.
   [U, A, An] [4]
9. Using a suitable computational software obtain the vibrational frequencies, bond length, bond angle, dipole moment & Total energy of CO_2 molecule.
   [U, A, An] [4]
10. Using a suitable computational software compare the IR and Raman spectra of H_2O molecule [U, A, An] [4]
11. Using a suitable computational software compare the IR and Raman spectra of CO_2 molecule [U, A, An] [4]
12. Determine Planck's constant using LEDs. Also do a graphical comparison. [U, A, An] [6]
13. Plot the V-I characteristics of TWO different LEDs and compare their turn-on voltages.

[U, A, An] [6]

14. Plot the V-I characteristics of a LED and determine the turn on voltage, ac and dc resistances. [U, A, An] [6]
15. Determine the wavelength of a laser using diffraction grating. [U, A, An] [6]
16. Plot the FTIR spectrum from the given data. Analyse it and give your interpretation. [U, A, An] [6]
17. Plot the Raman spectrum from the given data. Analyse it and give your interpretation. [U, A, An] [6]
18. Find the refractive indices of crown glass and flint glass prisms. Light source and spectrometer are given. [U, A, An] [6]